

Research Statement

Junrong Yan

My research lies at the interface of analysis, geometry, topology, and mathematical physics. A central question guiding my work is: *how can techniques from global analysis and index theory be used to understand spectral problems and to construct and compare geometric and topological invariants?* A recurring theme in my work is to convert global or singular problems into local geometric analysis. Witten deformation concentrates spectral information near critical sets, heat-kernel analysis connects local geometry with global spectral invariants, and compactifications of configuration spaces provide a geometric framework for analyzing the singularities in Feynman graph integrals. My research develops along three interconnected directions:

- (a) Feynman graph integrals in holomorphic quantum field theory and their applications to higher-genus B-model invariants;
- (b) analytic torsion and higher Cheeger-Müller theory, using Witten deformation to compare analytic and topological invariants;
- (c) spectral theory of Schrödinger-type operators on noncompact manifolds, motivated in part by Landau-Ginzburg models.

These directions come together in my long-term program toward a higher-genus Calabi-Yau/Landau-Ginzburg correspondence on the B-model side.

1 Overview of my results

Feynman graph integrals and holomorphic field theory. With Minghao Wang, we proved the convergence of Feynman graph integrals on closed real-analytic Kähler manifolds in the sense of Cauchy principal value [22]. On Calabi-Yau threefolds, this gives a direct and rigorous realization of the candidate higher-genus B-model invariants proposed by Bershadsky-Cecotti-Ooguri-Vafa. We subsequently proved that three natural rigorous renormalization procedures give the same answer and derived a gauge anomaly formula [23]. Such agreement is not generally expected in QFT, where different renormalization schemes may produce different finite parts.

Analytic torsion and higher torsion. Using Witten deformation by specially chosen non-Morse functions, I developed a localization method adapted to decompositions of manifolds. This gives a new purely analytic proof of the gluing formula for analytic torsion and, for families, a simpler proof of the gluing formula for Bismut-Lott torsion forms [24, 25]. With Martin Puchol [19], we proved a comparison of analytic and topological higher torsions for unitary flat bundles using generalized Morse functions. This removes the fiberwise Morse-function existence assumption in [3] and trivial flat bundle assumption in [20], while retaining Thom-Smale transversality. The proof resolves the main analytic difficulty caused by birth-death singularities.

Spectral theory on noncompact manifolds. With Maxim Braverman and Xianzhe Dai [8], we obtained an intrinsic criterion for the classical Weyl law for Schrödinger operators on general complete manifolds, removing both bounded geometry and the doubling condition from our earlier result [12]. We also constructed examples showing that the Weyl law can fail when the criterion is violated [8]. More recently, I proved that the weighted Laplacian on every complete gradient Ricci shrinker satisfies the classical Weyl law [26]. My earlier work with Dai established Agmon estimates, a comparison of the analytic Witten complex with a finite-dimensional complex generated by critical points, heat-kernel expansions and a local index theorem [10, 11].

2 Feynman graph integrals and holomorphic field theory

Background and question. In perturbative quantum field theory, Feynman graph integrals encode the contributions associated with interactions of particles. In mathematics, they also provide a striking bridge between physical theories and geometric or topological invariants. In topological quantum field theory (TQFT), such integrals converge in the Lebesgue sense and give rise to important topological invariants [1, 7, 17].

In general QFTs, however, graph integrals require renormalization, and different procedures may produce different values. Our work asks whether the geometry of holomorphic field theories on Kähler manifolds permits a canonical construction.

For a graph Γ with vertex set Γ_0 and edge set Γ_1 , assign a point $x_v \in M$ to each vertex. Each edge $e = (v, w)$ contributes a propagator $P_e(x_v, x_w)$, constructed from a Green kernel and its derivatives. The graph integral has the schematic form

$$\int_{M^{\Gamma_0}} \prod_{e=(v,w) \in \Gamma_1} P_e(x_v, x_w),$$

where $M^{\Gamma_0} = \prod_{v \in \Gamma_0} M$. The propagators are singular when their endpoints coincide (i.e., $x_v \rightarrow x_w$). The main analytic difficulty is to control simultaneous and nested collisions of vertices.

Convergence. In joint work with Wang, we proved the convergence of these graph integrals on closed real-analytic Kähler manifolds [22]. Using heat-kernel techniques together with Getzler rescaling from local index theory, we show that the graph integrands extend canonically to the Fulton-MacPherson compactification of configuration spaces (the compactification records how points collide) as forms with controlled singularities and therefore admit well-defined Cauchy principal-value regularizations.

Applications and significance of our results. When specialized to Calabi-Yau threefolds, this gives a direct and mathematically rigorous construction of candidate higher-genus ($g \geq 2$) B-model amplitudes through the Feynman graph integrals proposed by BCOV in [2]. In complex dimension one, earlier work of Li [18] and Costello-Li [9] constructed a punctured version of BCOV theory for elliptic curves whose large-complex-structure limit recovers punctured Gromov-Witten invariants of the mirror. This provides an important rigorous higher-genus instance of the mirror-symmetry picture underlying the BCOV construction.

Independence of the renormalization scheme. A further question is whether different renormalization procedures produce the same value. With Wang, we proved that heat-kernel renormalization, zeta regularization, and Cauchy principal-value regularization all give the same value for holomorphic field theories on closed real-analytic Kähler manifolds [23]. The difficulty is that these procedures organize the singularities in very different ways. We resolve these by constructing a wonderful compactification of the product of the configuration space with the complexified time parameters, allowing the three procedures to be compared.

We also derived a gauge anomaly formula: the failure of the resulting graph currents to be $\bar{\partial}$ -closed is expressed by boundary integrals over the compactified space of time parameters [23, §6]. This anomaly formula computes the obstruction to the quantum master equation in the BV formalism.

Future directions. We will next extend the construction to families and study anomaly formulas for Feynman graph integrals under variations of Kähler and complex structures. The goal is to understand the differential forms on moduli spaces of Calabi-Yau manifolds obtained from fiberwise Feynman graph integrals, in analogy with the analytic torsion forms discussed in §3. The main analytic problem is to obtain estimates on the propagators and their singularities that are uniform in the family parameter. A second, longer-term program is field theory on manifolds with boundary, where my Witten-deformation gluing methods may be used to separate bulk and boundary contributions.

3 Higher Cheeger-Müller/Bismut-Zhang theory

Background. A classical problem in global analysis is to understand why spectral invariants of differential operators can coincide with invariants defined by topological or combinatorial methods. Ray-Singer analytic torsion is a central example: it is defined from regularized determinants of Hodge Laplacians, whereas Reidemeister torsion is constructed from a finite-dimensional cellular complex. The Cheeger-Müller theorem identifies analytic and Reidemeister torsion. Bismut-Zhang extended the comparison to a more general setting using Witten deformation, which deforms the de Rham differential by a Morse function so that small eigenforms localize at its critical points [6].

For families of manifolds, the corresponding problem is to compare differential-form-valued analytic torsion with higher topological torsion over the parameter space. Let $\pi : M \rightarrow S$ be a smooth fibration with fibers $Z_s = \pi^{-1}(s)$. Bismut and Lott generalized the numerical analytic torsion of a single manifold to an even differential form for a flat complex vector bundle $F \rightarrow M$

$$\mathcal{T}_{\text{BL}}(M/S, F)$$

on the parameter space S [5]. Its degree-zero component recovers ordinary analytic torsion. On the topological side, Igusa and Klein constructed the topological analogue of higher analytic torsion, which can detect exotic smooth structures that are invisible to ordinary characteristic classes [13–15].

What was known and its limitations. A higher Cheeger-Müller/Bismut-Zhang theorem should identify the higher analytic torsion of Bismut-Lott with its topological counterpart. There are two main approaches to such a theorem, but both impose different and quite strong restrictions. Bismut-Goette’s analytic proof [3] applies to general flat bundles, but requires a fiberwise Morse function satisfying fiberwise Thom-Smale transversality (a genericity condition making the Thom-Smale complex well defined). The axiomatic approach in Puchol-Zhang-Zhu [20] removes the existence assumption on fiberwise Morse functions, but is restricted to trivial flat bundles.

Difficulties and result obtained. With Martin Puchol, we prove the comparison for general unitary flat bundles without assuming the existence of a fiberwise Morse function [19], thereby substantially removing the restrictions inherent in both approaches above. The axiomatic results in [20] reduce the theorem to a relative comparison. On the analytic side, the relative higher torsion is

$$\mathcal{T}_{\text{BL}}^{\text{rel}}(M/S, F) := \mathcal{T}_{\text{BL}}(M/S, F) - \mathcal{T}_{\text{BL}}(M/S, \underline{\mathbb{C}}^{\text{rk}(F)}),$$

where $\underline{\mathbb{C}}^{\text{rk}(F)}$ is the trivial flat bundle of the same rank as F ; the topological torsion is normalized in the same way.

The main difficulty is that this requires a new Witten-deformation analysis for fiberwise generalized Morse functions. Such functions always exist, but they may develop birth-death points, where the local form contains a cubic term such as x^3 , rather than only quadratic terms. As one approaches a fiber containing such a point, two critical points merge and the Hessian degenerates, so the estimates underlying the usual Morse analysis cease to be uniform. Establishing uniform estimates and controlling the resulting small eigenspaces therefore requires a substantial amount of delicate and technically demanding analysis.

We handle this degeneration by a localization argument adapted from the non-Morse Witten deformation I developed for torsion gluing formulas [24, 25], which isolates the contribution of the birth-death region. Since analytic torsion and torsion forms are defined through the heat kernels of the relevant operators, the analysis naturally separates into small- and large-time parts. In the small-time analysis, the isolated birth-death contribution cancels in the relative torsion against the corresponding contribution from the trivial bundle.. For large time, we combine Bismut-Lebeau’s techniques [4] with the noncompact Agmon estimates developed in my earlier joint work with Dai [10] to obtain the uniform estimates needed near the birth-death locus. Together with the axiomatic reduction above, this yields the higher Cheeger-Müller/Bismut-Zhang theorem for the higher torsions, while still *assuming fiberwise Thom-Smale transversality*.

Future direction: removing transversality by higher structures. My next goal is to remove global Thom-Smale transversality from this comparison theorem, paving the way toward a higher Cheeger-Müller/Bismut-Zhang theorem in full generality. Since the Thom-Smale condition can always be achieved locally on the base, I plan to choose vertical metrics satisfying it over a sufficiently fine open cover $S = \bigcup_i U_i$ and glue the resulting local Morse complexes by higher continuation data. On a double intersection $U_i \cap U_j$, two choices of metric are related by a continuation map. On a triple intersection $U_i \cap U_j \cap U_k$, the two possible compositions agree only up to chain homotopy, and on higher intersections one must introduce higher homotopies satisfying the corresponding coherence relations.

This higher structure should define a Čech version of the relative topological torsion, while local Witten deformation should identify it with the Bismut-Lott analytic torsion form. The new analytic issue is to control continuation maps and higher homotopies uniformly near birth-death loci, extending the estimates used in our previous comparison theorem.

4 Analysis of Schrödinger-type operators on noncompact manifolds

A basic problem in spectral geometry is to understand spectral invariants of Schrödinger-type operators on noncompact manifolds. This question also arises naturally in the Landau-Ginzburg B-model. An LG model consists of such a manifold X and a holomorphic function f , called the superpotential. The associated operator on differential forms has the form

$$\Delta_f = \Delta + |\nabla f|^2 + \text{a zero-order term involving } \text{Hess}(f).$$

Regularized determinants of this operator are expected to govern the genus-one theory.

A major obstacle in this program is that many fundamental tools of spectral geometry rely heavily on compactness and do not directly extend to noncompact manifolds with potentials. In a series of joint works with X. Dai, we developed analytic foundations for Schrödinger-type operators on noncompact manifolds, which are needed to define higher-genus BCOV invariants in the LG B-model. We proved exponential decay of eigenfunctions near infinity (Agmon estimates), which leads to the well-definedness of Thom-Smale-Witten cohomology for Morse functions on noncompact manifolds [10]. We also derived heat-kernel asymptotic expansions for such operators [11], allowing us to extend the local index theorem to noncompact manifolds with potentials. Lastly, defining zeta-regularized determinants requires control of the eigenvalue growth, which is exactly what the Weyl law provides.

Weyl law for Schrödinger operators on noncompact manifolds. A basic question in spectral geometry is whether the classical Weyl law continues to hold for Schrödinger operators on noncompact manifolds, where both the geometry at infinity and the growth of the potential can affect the eigenvalue distribution.

Using the heat-kernel expansion developed in [11], we established in [12] a *non-semiclassical Weyl law* for Schrödinger operators on noncompact manifolds with bounded geometry and potentials satisfying a doubling condition. This gives a Weyl law in a broad noncompact geometric setting without prescribing a specific model for the geometry at infinity. Even in special cases, such as hyperbolic manifolds with rapidly growing potentials studied in [16], which fall within the scope of our result, the analysis is already delicate.

In subsequent work with Braverman and Dai, we removed both the bounded geometry and doubling assumptions [8]. Our intrinsic criterion balances local curvature and injectivity radius against the growth and oscillation of the potential. It allows geometric irregularity at infinity, provided that this balance is preserved. We also constructed examples in which the Weyl law fails when the balance breaks down.

More recently, I applied these ideas to show that the weighted Laplacian on every complete gradient Ricci shrinker satisfies the classical Weyl law [26]. The difficulty is that, although the criterion in [8] is very general, it still cannot be applied directly to Ricci shrinkers. The key observation is that, roughly speaking, the problematic locus where this criterion may fail is asymptotically negligible. To establish this, I prove a large-scale regularity estimate showing that, in geodesic balls of radius R , the region with curvature radius below R^{-1} occupies a vanishing proportion of the volume as $R \rightarrow \infty$.

My next goal is to go beyond the leading Weyl asymptotics and obtain quantitative remainder estimates. On compact manifolds, refined estimates are derived from the microlocal structure of the wave kernel and Fourier Tauberian arguments, and are closely related to the dynamics of the geodesic flow. Extending these techniques to Schrödinger operators on noncompact manifolds would require controlling simultaneously the Hamiltonian dynamics and the geometry at infinity.

5 Toward a higher-genus Calabi-Yau/Landau-Ginzburg correspondence on the B-side

My long-term goal is a higher-genus Calabi-Yau/Landau-Ginzburg (CY/LG) correspondence on the B-model side, relating a compact Calabi-Yau theory to a noncompact LG model. The theory is organized by genus. At genus zero, the B-model is governed by tt^* -geometry, which can be viewed as a generalized variation of Hodge structures. With Xinxing Tang, we proved the CY/LG correspondence for Weil-Petersson metrics and tt^* -structures [21].

Genus one. My immediate goal is a genus-one comparison for Calabi-Yau threefolds. On the LG side, I am studying a modified analytic torsion built from regularized determinants of the deformed Laplacian. In ongoing work, I have derived its anomaly formula under variations of the superpotential and compared it with the anomaly formula for BCOV torsion on Calabi-Yau threefolds. This work is currently being prepared for publication. Agreement of the anomaly formulas would not by itself identify the invariants: a further task is to determine the remaining ambiguity through normalization and moduli boundary asymptotics. Our genus-zero identification of Weil-Petersson metrics and tt^* -structures provides a natural starting point for this comparison.

Higher genus. For $g \geq 2$, the next problem is to combine the graph-integral construction of §2 with the noncompact spectral analysis of §4. On the compact CY side, our graph integrals define candidate higher-genus B-model invariants. On the LG side, a new difficulty arises because vertices can collide while simultaneously escaping to infinity. Estimates near collision diagonals and estimates at infinity must therefore remain compatible when both phenomena occur together.

My approach is to construct a compactification of the configuration space that records both collision diagonals and escape to infinity, and to establish propagator estimates that remain uniform near their intersections. The compactification should distinguish the two different degeneration patterns so that their analytic behavior can be studied. The main task is to determine the asymptotics of the propagators near each boundary stratum and prove that the resulting graph integrals admit a canonical regularization.

This would provide a rigorous construction of candidate higher-genus B-model invariants on the LG side, making their comparison with the compact CY construction a precise mathematical problem.

Together with the genus-one torsion comparison, this program brings together the main techniques developed in my work: configuration-space compactifications for collision singularities, heat-kernel and spectral estimates for noncompact Schrödinger operators, and localization methods associated with the superpotential. The ultimate goal is to construct the higher-genus invariants on both sides within compatible analytic frameworks and prove their correspondence.

References

- [1] S. Axelrod and I. M. Singer. Chern-Simons perturbation theory. II. *Journal of Differential Geometry*, 39(1):173–213, 1994.
- [2] M. Bershadsky, S. Cecotti, H. Ooguri, and C. Vafa. Kodaira-Spencer theory of gravity and exact results for quantum string amplitudes. *Communications in Mathematical Physics*, 165:311–427, 1994.
- [3] J.-M. Bismut and S. Goette. *Families Torsion and Morse Functions*, volume 275 of *Astérisque*. Société Mathématique de France, 2001.

- [4] J.-M. Bismut and G. Lebeau. Complex immersions and Quillen metrics. *Publications Mathématiques de l'IHÉS*, 74:1–298, 1991.
- [5] J.-M. Bismut and J. Lott. Flat vector bundles, direct images and higher real analytic torsion. *J. Amer. Math. Soc.*, 8:291–363, 1995.
- [6] J.-M. Bismut and W. Zhang. *An Extension of a Theorem by Cheeger and Müller*, volume 205 of *Astérisque*. Société Mathématique de France, 1992.
- [7] R. Bott and A. S. Cattaneo. Integral invariants of 3-manifolds. *Raoul Bott: Collected Papers*, 5:383, 2018.
- [8] M. Braverman, X. Dai, and J. Yan. The classical Weyl law for Schrödinger operators on complete riemannian manifolds. *arXiv:2605.07200*, 2026.
- [9] K. Costello and S. Li. Quantum BCOV theory on Calabi-Yau manifolds and the higher genus B-model. *arXiv:1201.4501*, 2012.
- [10] X. Dai and J. Yan. Witten deformation for noncompact manifolds with bounded geometry. *J. Inst. Math. Jussieu*, 22:2095–2132, 2023.
- [11] X. Dai and J. Yan. Witten deformation on non-compact manifold: Heat kernel expansion and local index theorem. *Math. Z.*, 303(1), 2023.
- [12] X. Dai and J. Yan. Weyl law for Schrödinger operators on noncompact manifolds, heat kernel, and Karamata–Hardy–Littlewood theorem. *arXiv:2504.15551*, 2025.
- [13] S. Goette and K. Igusa. Exotic smooth structures on topological fiber bundles ii. *Transactions of the American Mathematical Society*, 366(2):791–832, 2014.
- [14] S. Goette, K. Igusa, and B. Williams. Exotic smooth structures on topological fiber bundles i. *Transactions of the American Mathematical Society*, 366(2):749–790, 2014.
- [15] K. Igusa. *Higher Franz-Reidemeister torsion*. Number 31. American Mathematical Soc., 2002.
- [16] Y. Inahama and S.-I. Shirai. Eigenvalue asymptotics for the Schrödinger operators on the real and the complex hyperbolic spaces. *Journal de Mathématiques Pures et Appliquées*, 83(5):589–627, 2004.
- [17] M. Kontsevich. Feynman diagrams and low-dimensional topology. In *First European Congress of Mathematics Paris, July 6–10, 1992*, pages 97–121. Springer, 1994.
- [18] S. Li. BCOV theory on the elliptic curve and higher genus mirror symmetry. *arXiv:1112.4063*, 2011.
- [19] M. Puchol and J. Yan. Generalized Morse functions, excision and higher torsions. *arXiv:2407.17100*, 2024.
- [20] M. Puchol, Y. Zhang, and J. Zhu. Additivity axiom for the torsion forms and higher Cheeger–Müller/Bismut–Zhang theorem. *Journal of the Institute of Mathematics of Jussieu*, 25(4):1833–1920, 2026.
- [21] X. Tang and J. Yan. CY/LG correspondence for Weil–Petersson metrics and tt^* structures. *J. Geom. Phys.*, 216:105545, 2025.
- [22] M. Wang and J. Yan. Feynman graph integrals on Kähler manifolds. *arXiv:2507.09170*, 2025.
- [23] M. Wang and J. Yan. Renormalizations in holomorphic field theories on Kähler manifolds. *arXiv:2608.00546*, 2026.
- [24] J. Yan. A new analytic proof of the gluing formula for analytic torsion forms. *arXiv:2301.02591*, 2023.
- [25] J. Yan. Witten deformation for non-Morse functions and gluing formula for analytic torsions. *Comm. Anal. Geom.*, to appear; *arXiv:2301.01990*, 2023.
- [26] J. Yan. The Weyl law meets large-scale regularity on Ricci shrinkers. *arXiv:2609.08206*, 2026.